

QUAID-I-AZAM UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE (BS 5th)

Terminal Examination Fall, 2020

Course No. ST-101 Course Title: Probability and Statistics

Max. Marks: 50

Time Allowed: 3 Hours

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Date: 15-JAN-2021

Note: Please attempt five questions. All questions carry equal marks.

Q.1 (a) What is meant by skewness? Distinguish between positive and negative skewness? (04)

Calculate the Bowley's co-efficient of skewness from the following distribution. (06)

Ages (years)	15-19	20-24	25-29	30-34	35-39	40-44	45-49	50-54
No. of Men	29	176	208	173	82	40	15	3

Q.2 (a) What is meant by conditional probability? Briefly explain Simple and Compound Events? (03)

(b) A professor of statistics has two boxes of computer disks: box C_1 contains seven Verbatim disks and three Control Data disks and box C_2 contains two Verbatim disks and eight Control Data disks. She selects a box at random with probabilities $P(C_1) = 2/3$. And $P(C_2) = 1/3$ because of their respective locations. A disk is then selected at random and the event C occurs if it is from Control Data, Using an equally likely assumption for each disk in the selected box, compute $P(C_1 | C)$ and $P(C_2 | C)$. (03)

(c) Three distinct integers are chosen at random from the first 20 positive integers. Compute probability that: (i) their sum is even; (ii) their product is even (04)

Q.3 (a) Explain clearly the concepts of Random Variable. What is a distribution function? (03)

(b) The following table shows the joint probability distribution of random variables X and Y (04)

$Y \backslash X$	1	2
1	2/15	1/15
3	4/15	1/15
5	3/15	4/15

Compute the conditional distribution $f(x/y)$. Check whether X and Y are independent random variables.

(c) For what value of A , the function defined as below will be a p.d.f (03)

$$f(x) = Ax^3(1-x), \quad 0 \leq x \leq 1$$

$$= 0 \quad \text{elsewhere}$$

Also find its mean and variance.

Q.4 (a) Describe a Poisson Distribution with the help of at least two real life examples? (02)

(b) According to national fire protection report more than one-third of all hotel fires are caused by guest room furnishing. Suppose that guest room furnishings caused a hotel fire is 35%. What is the probability that three of next four hotel fires will be caused by the guest room furnishings, assuming independence of events? (04)

(c) Given that the height of college boys is normally distributed with mean 5'-2" and standard deviation 4", and that the minimum height required for joining the N.C.C. is 5'-4"; find the percentage of boys who would be rejected on account of their height. (04)

Q.5 (a) How could you define Normal Distribution in relation with its parameters? (02)

(b) Heights of applicants to police force are normally distributed with mean 170 cm and S.D 3.8 cm. If 30% of applicants are rejected because they are too small, what is minimum acceptable height for police force? (04)

(c) Three urns of the same appearance have the following proportions of white and black balls. (04)

Urn A: 1 white, 2 black balls

Urn B: 2 white, 1 black balls

Urn C: 2 white, 2 black balls

One of the urn is selected and a ball is drawn from it. It turns out to be white, what is the probability that urn C was chosen?

BEST OF LUCK